

Tasnia Chowdhury

Computer Science Student · Full-Stack Developer · ML Researcher

Premier University, Chattogram



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PROFILE

Final-year Computer Science undergraduate with hands-on experience building full-stack web applications, graphics designs, branding, designing RESTful APIs, and applying machine learning to real-world NLP problems. Proven ability to lead small development teams and ship production-ready projects. Passionate about ML, multilingual NLP, and impactful technology for underserved communities.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Java, C, SQL
Frontend	React, Vite, Tailwind CSS, DaisyUI, HTML/CSS
Backend	Node.js, Express, FastAPI, REST APIs, Firebase Admin SDK
Databases	MongoDB Atlas, PostgreSQL, Firebase Firestore
Cloud & DevOps	Vercel, Railway, Netlify, GitHub Actions
ML / NLP	Federated Learning, Hugging Face Transformers, scikit-learn
Networking	Cisco Packet Tracer, TCP/IP, VLANs, RIPv2, DHCP, DNS
Tools	Git, Postman, Figma, Adobe Illustrator, Adobe Photoshop

PROJECTS

HappyPaws— Animal Welfare Donation Platform

React · Node.js · Vercel

- Built a full-stack donation and adoption platform with pet listings, an admin dashboard, and role-based access control.
- Integrated SSLCommerz payment gateway, Cloudinary media management, and Firebase authentication; deployed serverless API routes on Vercel.
- Led a small team using a structured GitHub branching workflow; resolved complex CORS and MongoDB Atlas networking issues in production.

FastAPI Public Pet API

Python · FastAPI · GitHub Actions CI/CD

- Developed a public REST API for pet data; built and debugged automated CI/CD
- pipeline with GitHub Actions.

EasySave- Budget Tracker

React · Node.js · Vercel

ISP Network Topology (Cisco Packet Tracer)

Cisco Packet Tracer

Designed a full ISP topology for a Chittagong-based network with IIG routers, inter-VLAN routing, and complete server-service stack.

Smart Irrigation System

ESP32 · IoT Sensors · Mobile App Integration

- Developed an automated irrigation system using soil moisture, pH, and temperature sensors to monitor field conditions in real time.
- Integrated ESP32 microcontroller with mobile app for remote monitoring and control of irrigation cycles.
- Enabled crop-specific water level adjustments, reducing water waste while optimising crop health.
- Published a conference paper on my IOT project.

AI Malicious URL Detection

Python · Machine Learning · NLP

- Built a machine learning model to classify URLs as malicious or benign using lexical and host-based features.
- Applied NLP-based feature extraction and evaluated multiple classifiers to maximise detection accuracy with low false-positive rates.

Book Exchange & Course Information Website

Web Development · React · Node.js

- Developed a web platform enabling students to list, share, and exchange academic books within their university community.
- Aggregated course information from multiple universities into a unified system for easy student reference.
- Designed with a focus on usability and accessibility to serve diverse student communities effectively.

EDUCATION

B.Sc. in Computer Science & Engineering

Premier University Class of 2027

Chittagong, Bangladesh

- Relevant coursework: Computer Networks, Machine Learning, Database Systems, Software Engineering, Algorithms & Data Structures.

ADDITIONAL

Languages

Bangla (native)

English (fluent)

Interests

Graphic design & logo work · Open-source development · Animal welfare technology